

Week 3

Wednesday 1/28

consistency: a numerical method is consistent w/ IVP if

$$\star \quad \lim_{h \rightarrow 0} \tau_{i+1}(h) = 0$$

$$\tau_{i+1}(h) = \frac{y_{i+1} - y_i}{h} - f(t_i, y_i)$$

→ addresses the error in using a difference eq to approx the differential eqn.

We want to understand how τ_{i+1} behaves as h changes

$$\tau_{i+1}(h) = \frac{y_{i+1} - y_i}{h} - f(t_i, y_i) = \frac{y_i + h y_i' + \frac{h^2}{2} y_i'' + O(h^3) - y_i}{h} - \varphi(t_i, y_i) \quad \text{where } \varphi = \frac{w_{i+1} - w_i}{h}$$

$$= \underbrace{y_i' - f(t_i, y_i)}_{\text{should be 0}} + \frac{h}{2} y_i'' + O(h^2)$$

ex. Show the method is not consistent w/ IVP $y' = f(t, y)$, $y(t_0) = y_0$

$$w_{i+1} = w_i + h \underbrace{(f(t_i, w_i) + f(t_{i+1}, w_{i+1}))}_{\varphi}$$

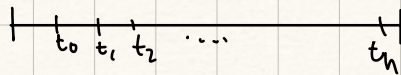
$$\begin{aligned} \tau_{i+1} &= \frac{y_{i+1} - y_i}{h} - \varphi(t_i, y_i) \\ &= \frac{y_i + h y_i' + \frac{h^2}{2} y_i'' + O(h^3) - y_i}{h} - \varphi(t_i, y_i) \\ &= y_i' - f(t_i, y_i) + \frac{h}{2} y_i'' + O(h^2) \\ &= y_i' - (f_i + f_{i+1}) + \frac{h}{2} y_i'' + O(h^2) \\ &= y_i' - (y_i' + y_{i+1}') + \frac{h}{2} y_i'' + O(h^2) \\ &= -y_{i+1}' + \frac{h}{2} y_i'' + O(h^2) \\ &= -(y_i' + h y_i'' + O(h^2)) + \frac{h}{2} y_i'' + O(h^2) \\ &= -y_i' - \frac{h}{2} y_i'' + O(h^2) \Rightarrow \lim_{h \rightarrow 0} \tau_{i+1}(h) \neq 0 \end{aligned}$$

as $h \rightarrow 0$, we get $-y_i'$

∴ the method is not consistent

A method is worthless unless it converges

def A numerical method converges if $\lim_{h \rightarrow 0} [\max_i |w_i - y_i|] = 0$



def stability concerns the error due to round off
 → sensitivity to perturbations

Runge - Kutta

GOAL: determine methods of the form

$$w_{i+1} = w_i + h \varphi(t_i, w_i)$$

→ called "one-step" method

→ that are higher order & do not require calculation of higher order derivatives

$$\rightarrow \frac{dy}{dt} = f(t, y)$$

$$\rightarrow y(t_i + h) = y(t_i) + \int_{t_i}^{t_i+h} f(\tau, y(\tau)) d\tau \quad (\text{fundamental thm of calculus})$$

let's think about quadrature methods like Gaussian Quadrature:

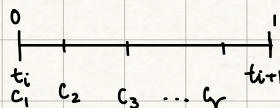
replace integral with

... first, quick sub to clean...

$$\tau = t_i + ht$$

$$\int_0^1 f(t_i + ht, y(t_i + ht)) dt \approx \sum_{j=1}^r b_j f(t_i + hc_j, y(t_i + hc_j))$$

don't know... approx these



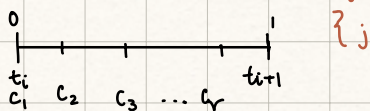
Friday 11/30

idea: integrate the ODE

$$y_{i+1} = y_i + \int_{t_i}^{t_{i+1}} f(\tau, y(\tau)) d\tau$$

$$= y_i + \int_0^1 f(t_i + ht, y(t_i + ht)) dt$$

$$\approx y_i + \sum_{j=1}^r b_j f(t_i + c_j h, y(t_i + c_j h))$$



note: let $C_1 = 0$, remaining C_j 's are free

$$y(t_i, c_j h) = y_i + \int_{t_i}^{t_i + c_j h} f(t, y(t)) dt$$

(fundamental thm of calculus)

$$z_i = y_i$$

$$z_2 = y_1 + h\alpha_2 f(t_1, z_1)$$

$$z_3 = y_i + h a_{31} f(t_i, z_1) + h a_{32} f(t_i + c_2 h, z_2)$$

;

$$z_v = y_i + h \sum_{k=1}^{v+1} a_{v,k}(t_i + c_k h, z_k)$$

2 quadrature methods

Runge Kutta Tableau

lower triangular matrix

$$O = \begin{pmatrix} c_1 & & & & \\ & c_2 & & & \\ & & \ddots & & \\ & & & \ddots & \\ & & & & c_j \end{pmatrix} \quad \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1j} \\ a_{21} & a_{22} & \dots & a_{2j} \\ \vdots & \vdots & \ddots & \vdots \\ a_{j1} & a_{j2} & \dots & a_{jj} \end{pmatrix}$$

Set up
for explicit
RK